CYBERSRS

Email Security Gateway Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages. The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events. The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

FR-001 - protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Statement: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Rationale: Implements stakeholder source E-001.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: An email containing a phishing link is sent to a university user

When: The email security gateway processes the incoming message

Then: The system detects the phishing content and blocks delivery

Method: test

Observable evidence: Test capture of the blocked message with phishing indicator

Given: An email with a malicious attachment is sent to a university user

When: The email security gateway processes the incoming message

Then: The system identifies the malicious attachment and quarantines it

Method: test

Observable evidence: Test capture of the quarantined file with malware signature

Given: An email spoofed with a fake university domain is sent to a user

When: The email security gateway processes the incoming message

Then: The system flags the sender as spoofed and alerts administrators

Method: test

Observable evidence: Test capture of the spoofed header and alert notification

Given: An email with a suspicious link is sent to a user

When: The email security gateway processes the incoming message

Then: The system detects the suspicious URL and marks it for review

Method: test

Observable evidence: Test capture of the flagged link and user notification

Unresolved decisions: Thresholds for what constitutes a 'suspicious' link or attachment

FR-002 - inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Statement: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Rationale: Implements stakeholder source E-002.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: An incoming email with malicious content is received by the gateway

When: The system performs inspection on the message body and headers

Then: The system quarantines the message and generates a security alert

Method: test

Observable evidence: Test capture of quarantine action and generated alert

Given: An outgoing email is sent from a university user to an external recipient

When: The system inspects the content for malicious payloads or links

Then: The system detects suspicious content and blocks transmission

Method: test

Observable evidence: Test capture of blocked outbound message with reason

Given: A quarantined email is identified by an administrator

When: An authorized admin requests review and release of the message

Then: The system allows the administrator to view and approve or reject the message

Method: test

Observable evidence: Test capture of admin interface response for quarantined email

Given: A suspicious message is detected during processing

When: The system generates a security alert

Then: The alert is delivered to designated administrators via secure channel

Method: test

Observable evidence: Test capture of alert delivery to admin console

Unresolved decisions: Authentication method for administrator access to quarantined messages; Definition of 'authorized' administrative role

3. Non-functional requirements

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

NFR-001 - The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Statement: The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Rationale: Implements stakeholder source E-004.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A system administrator initiates a deployment request for university-wide use

When: The system processes the deployment request

Then: The system returns a status indicating successful validation of security, reliability, scalability, and manageability criteria

Method: test

Observable evidence: Test capture of the system's response to deployment request with validated criteria

Given: A network operations team observes increasing load during peak hours

When: The system handles traffic under sustained high-load conditions

Then: The system maintains service availability and performance without degradation

Method: inspection

Observable evidence: Test capture of system performance metrics during peak load

Given: A new department requests integration with existing university services

When: The system is configured to support new service connections

Then: The system enables configuration without requiring revalidation or manual intervention

Method: demonstration

Observable evidence: Test capture of configuration workflow and successful service integration

Unresolved decisions: Definition of 'manageable' in context of university-wide deployment; Thresholds for scalability under concurrent user load; Validation method for reliability during failure scenarios

4. Security, privacy and compliance

Applicability reason: Source-backed obligations; unspecified decisions remain open.

Requirements

SEC-001 - The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Statement: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Rationale: Implements stakeholder source E-003.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: An administrator performs a filtering action on an email message

When: The system records the filtering decision and associated user action

Then: The audit log contains a record of the filtering decision and the administrator's identity and timestamp

Method: test

Observable evidence: Test capture of the audit log entry

Given: An email security event occurs in the system

When: The platform generates an alert for monitoring via its dashboard

Then: The dashboard displays the event with relevant metadata including source, severity, and timestamp

Method: inspection

Observable evidence: Test capture of dashboard event display

Given: The university identity system is integrated with the platform

When: An administrator logs in using their university credentials

Then: The platform authenticates the user and grants access based on role and permissions

Method: demonstration

Observable evidence: Test capture of authentication and access grant

Unresolved decisions: Definition of 'filtering decision' in audit logs; Threshold for event severity in dashboard alerts; Authentication method used during integration with identity system

Related requirement ids: FR-001; FR-002

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: SEC-001

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

7. Compliance and policy applicability matrix

Open decisions: How to define 'dynamic access policies' for email traffic based on user behavior and device context; What thresholds trigger a 'suspicious' link or attachment in phishing detection; Whether authentication methods for administrator access should be multi-factor; Definition of ' manageable' in the context of university-wide deployment; Thresholds for scalability under concurrent user load; Validation method for reliability during failure scenarios; How to define 'filtering decision' in audit logs; Threshold for event severity in dashboard alerts; Authentication method used during integration with identity system

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: authorized administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Component name: Actor: authorized administrators

Responsibility: Represent the source-stated actor: authorized administrators.

Email Security Engine

Detail: Proposed logical owner for FR-001, FR-002, NFR-001.

Status: proposed

Evidence ids: E-001; E-002; E-004

Requirement ids: FR-001; FR-002; NFR-001

Component name: Email Security Engine

Responsibility: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages. The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Audit and Monitoring Dashboard

Detail: Proposed logical owner for SEC-001, NFR-001.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: SEC-001; NFR-001

Component name: Audit and Monitoring Dashboard

Responsibility: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events. The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

From component: Actor: authorized administrators

To component: Email Security Engine

Data or action: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002

Untrusted component: Actor: authorized administrators

Trusted component: Email Security Engine

Enforcement: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Email Security Engine

Trusted component: Audit and Monitoring Dashboard

Enforcement: The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for NFR-001.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-001

Untrusted component: Audit and Monitoring Dashboard

Trusted component: Email Security Engine

Enforcement: The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: authorized administrators

Goal: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Story: As authorized administrators, I need the following behavior: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined

messages.

Preconditions

1. A quarantined email is identified by an administrator

Steps

1. An authorized admin requests review and release of the message

Success outcome: The system allows the administrator to view and approve or reject the message

Alternate flows

1. Given: An incoming email with malicious content is received by the gateway When: The system performs inspection on the message body and headers Then: The system quarantines the message and generates a security alert Evidence: Test capture of quarantine action and generated alert
2. Given: An outgoing email is sent from a university user to an external recipient When: The system inspects the content for malicious payloads or links Then: The system detects suspicious content and blocks transmission Evidence: Test capture of blocked outbound message with reason

Requirement ids: FR-002

Evidence ids: E-002

Unresolved decisions: Authentication method for administrator access to quarantined messages; Definition of 'authorized' administrative role

Id: UC-002

Actor: authorized administrators

Goal: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Story: As authorized administrators, I need the following behavior: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Preconditions

1. A suspicious message is detected during processing

Steps

1. The system generates a security alert

Success outcome: The alert is delivered to designated administrators via secure channel

Requirement ids: FR-002

Evidence ids: E-002

Unresolved decisions: Authentication method for administrator access to quarantined messages; Definition of 'authorized' administrative role

Open issues: Authentication method for administrator access to quarantined messages; Definition of ' authorized' administrative role

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: An email containing a phishing link is sent to a university user
2. When: The email security gateway processes the incoming message

Expected result: Expected: The system detects the phishing content and blocks delivery

Evidence to record: Evidence: Test capture of the blocked message with phishing indicator

Test type: positive

Id: TEST-002

Requirement ids: FR-001

Procedure

1. Given: An email with a malicious attachment is sent to a university user
2. When: The email security gateway processes the incoming message

Expected result: Expected: The system identifies the malicious attachment and quarantines it

Evidence to record: Evidence: Test capture of the quarantined file with malware signature

Test type: positive

Id: TEST-003

Requirement ids: FR-001

Procedure

1. Given: An email spoofed with a fake university domain is sent to a user
2. When: The email security gateway processes the incoming message

Expected result: Expected: The system flags the sender as spoofed and alerts administrators

Evidence to record: Evidence: Test capture of the spoofed header and alert notification

Test type: positive

Id: TEST-004

Requirement ids: FR-001

Procedure

1. Given: An email with a suspicious link is sent to a user
2. When: The email security gateway processes the incoming message

Expected result: Expected: The system detects the suspicious URL and marks it for review

Evidence to record: Evidence: Test capture of the flagged link and user notification

Test type: positive

Id: TEST-005

Requirement ids: FR-002

Procedure

1. Given: An incoming email with malicious content is received by the gateway
2. When: The system performs inspection on the message body and headers

Expected result: Expected: The system quarantines the message and generates a security alert

Evidence to record: Evidence: Test capture of quarantine action and generated alert

Test type: positive

Id: TEST-006

Requirement ids: FR-002

Procedure

1. Given: An outgoing email is sent from a university user to an external recipient
2. When: The system inspects the content for malicious payloads or links

Expected result: Expected: The system detects suspicious content and blocks transmission

Evidence to record: Evidence: Test capture of blocked outbound message with reason

Test type: positive

Id: TEST-007

Requirement ids: FR-002

Procedure

1. Given: A quarantined email is identified by an administrator
2. When: An authorized admin requests review and release of the message

Expected result: Expected: The system allows the administrator to view and approve or reject the message

Evidence to record: Evidence: Test capture of admin interface response for quarantined email

Test type: positive

Id: TEST-008

Requirement ids: FR-002

Procedure

1. Given: A suspicious message is detected during processing
2. When: The system generates a security alert

Expected result: Expected: The alert is delivered to designated administrators via secure channel

Evidence to record: Evidence: Test capture of alert delivery to admin console

Test type: positive

Id: TEST-009

Requirement ids: NFR-001

Procedure

1. Given: A system administrator initiates a deployment request for university-wide use
2. When: The system processes the deployment request

Expected result: Expected: The system returns a status indicating successful validation of security, reliability, scalability, and manageability criteria

Evidence to record: Evidence: Test capture of the system's response to deployment request with validated criteria

Test type: positive

Id: TEST-010

Requirement ids: NFR-001

Procedure

1. Given: A network operations team observes increasing load during peak hours
2. When: The system handles traffic under sustained high-load conditions

Expected result: Expected: The system maintains service availability and performance without degradation

Evidence to record: Evidence: Test capture of system performance metrics during peak load

Test type: inspection

Id: TEST-011

Requirement ids: NFR-001

Procedure

1. Given: A new department requests integration with existing university services
2. When: The system is configured to support new service connections

Expected result: Expected: The system enables configuration without requiring revalidation or manual intervention

Evidence to record: Evidence: Test capture of configuration workflow and successful service integration

Test type: positive

Id: TEST-012

Requirement ids: SEC-001

Procedure

1. Given: An administrator performs a filtering action on an email message
2. When: The system records the filtering decision and associated user action

Expected result: Expected: The audit log contains a record of the filtering decision and the administrator's identity and timestamp

Evidence to record: Evidence: Test capture of the audit log entry

Test type: positive

Id: TEST-013

Requirement ids: SEC-001

Procedure

1. Given: An email security event occurs in the system
2. When: The platform generates an alert for monitoring via its dashboard

Expected result: Expected: The dashboard displays the event with relevant metadata including source, severity, and timestamp

Evidence to record: Evidence: Test capture of dashboard event display

Test type: inspection

Id: TEST-014

Requirement ids: SEC-001

Procedure

1. Given: The university identity system is integrated with the platform
2. When: An administrator logs in using their university credentials

Expected result: Expected: The platform authenticates the user and grants access based on role and permissions

Evidence to record: Evidence: Test capture of authentication and access grant

Test type: positive

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to favor one SDLC model over others. All models receive equal suitability points (40) due to absence of factors such as budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. The decision remains provisional until these unknown factors are clarified.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.; FR-002: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.; SEC-001: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.
2. Review and baseline the source-defined work group: NFR-001: The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; NFR-001; SEC-001

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.; FR-002: The system shall inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.; SEC-001: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.
3. Implement the source-defined work group: NFR-001: The solution shall be secure, reliable, scalable, and manageable for university-wide deployment.
4. In an authorized isolated test environment, execute TEST-001: Given: An email containing a phishing link is sent to a university user When: The email security gateway processes the incoming message Expected: The system detects the phishing content and blocks delivery
5. In an authorized isolated test environment, execute TEST-002: Given: An email with a malicious attachment is sent to a university user When: The email security gateway processes the incoming message Expected: The system identifies the malicious attachment and quarantines it
6. In an authorized isolated test environment, execute TEST-003: Given: An email spoofed with a fake university domain is sent to a user When: The email security gateway processes the incoming message Expected: The system flags the sender as spoofed and alerts administrators
7. In an authorized isolated test environment, execute TEST-004: Given: An email with a suspicious link is sent to a user When: The email security gateway processes the incoming message Expected: The system detects the suspicious URL and marks it for review
8. In an authorized isolated test environment, execute TEST-005: Given: An incoming email with malicious content is received by the gateway When: The system performs inspection on the message body and headers Expected: The system quarantines the message and generates a security alert
9. In an authorized isolated test environment, execute TEST-006: Given: An outgoing email is sent from a university user to an external recipient When: The system inspects the content for malicious payloads or links Expected: The system detects suspicious content and blocks transmission
10. In an authorized isolated test environment, execute TEST-007: Given: A quarantined email is identified by an administrator When: An authorized admin requests review and release of the message Expected: The system allows the administrator to view and approve or reject the message
11. In an authorized isolated test environment, execute TEST-008: Given: A suspicious message is detected during processing When: The system generates a security alert Expected: The alert is delivered to designated administrators via secure channel
12. In an authorized isolated test environment, execute TEST-009: Given: A system administrator initiates a deployment request for university-wide use When: The system processes the deployment request Expected: The system returns a status indicating successful validation of security, reliability, scalability, and manageability criteria
13. In an authorized isolated test environment, execute TEST-010: Given: A network operations team observes increasing load during peak hours When: The system handles traffic under sustained

high-load conditions Expected: The system maintains service availability and performance withoutdegradation

14. In an authorized isolated test environment, execute TEST-011: Given: A new department requests integration with existing university services When: The system is configured to support new service connections Expected: The system enables configuration without requiring revalidation or manual intervention
15. In an authorized isolated test environment, execute TEST-012: Given: An administrator performs a filtering action on an email message When: The system records the filtering decision and associated user action Expected: The audit log contains a record of the filtering decision and the administrator's identity and timestamp
16. In an authorized isolated test environment, execute TEST-013: Given: An email security event occurs in the system When: The platform generates an alert for monitoring via its dashboard Expected: The dashboard displays the event with relevant metadata including source, severity, and timestamp
17. In an authorized isolated test environment, execute TEST-014: Given: The university identity system is integrated with the platform When: An administrator logs in using their university credentials Expected: The platform authenticates the user and grants access based on role and permissions
18. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
19. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; NFR-001; SEC-001

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.

2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against SEC-001: The platform shall maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; NFR-001; SEC-001

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Id: E-002

Source: description

Quote: The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Id: E-003

Source: description

Quote: The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Id: E-004

Source: description

Quote: The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Source references

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.7453

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kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

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Page or section: text chunk 2

Relevance score: 0.71

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Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7076

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kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7071

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:14 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7113

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

kb-0150d5352d0ec304:15 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 15

Page or section: text chunk 16

Relevance score: 0.7097

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-92ffa375cd05a8f5:19 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 19

Page or section: text chunk 20

Relevance score: 0.7107

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: qwen3:4b-instruct-2507-q4_K_M

Security, privacy and compliance: qwen3:4b-instruct-2507-q4_K_M

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.28

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, TEST-002, TEST-003, TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	UC-001, UC-002, TEST-005, TEST-006, TEST-007, TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-004	TEST-009, TEST-010, TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-001	E-003	TEST-012, TEST-013, TEST-014, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.28